

PostgreSQL Architecture Interview Questions and Answers

1. PostgreSQL Architecture

PostgreSQL follows a client-server architecture.

Main components:

- Client Application
- PostgreSQL Server
- Postmaster process
- Backend processes
- Shared memory
- Storage system

Postmaster:

Main server process that accepts connections and starts backend processes.

Backend Process:

Each client connection gets a separate process which executes SQL queries.

2. SQL Query Execution Flow

Flow:

Client -> Parser -> Analyzer -> Planner/Optimizer -> Executor -> Storage

Parser:

Checks SQL syntax.

Analyzer:

Checks tables, columns and permissions.

Planner:

Creates the best execution plan.

Executor:

Runs the query.

3. Memory Architecture

Shared Memory:

- Shared Buffers: Cache database pages.
- WAL Buffer: Stores transaction logs before writing.

Local Memory:

- work_mem: Used for sorting, joins and aggregation.
- maintenance_work_mem: Used for vacuum and index creation.

4. Storage Architecture

Structure:

Cluster
|
Database
|
Schema
|
Table
|
Page
|
Tuple

Page:

Default PostgreSQL page size is 8 KB.

Tuple:

A row stored inside a page.

5. WAL (Write Ahead Logging)

PostgreSQL writes changes to WAL before writing data files.

Benefits:

- Crash recovery
- Durability
- Replication

6. MVCC (Multi Version Concurrency Control)

PostgreSQL maintains multiple row versions.

Advantages:

- Readers do not block writers.
- Better concurrency.

Old row versions are removed by VACUUM.

7. VACUUM

VACUUM removes dead tuples created by MVCC.

Types:

- VACUUM
- VACUUM FULL

VACUUM ANALYZE:

Removes dead rows and updates optimizer statistics.

8. Index Architecture

Indexes improve query performance.

Default index:

B-Tree

Without index:

Full table scan

With index:

Index scan

9. Background Processes

Important processes:

Checkpoint:

Writes dirty pages to disk.

WAL Writer:

Writes WAL buffers.

Background Writer:

Writes changed pages.

Autovacuum:

Automatically cleans dead tuples.

Archiver:

Archives WAL files.

10. Replication Architecture

Primary database sends WAL to standby.

Types:

- Streaming replication
- Logical replication

11. Crash Recovery

Steps:

1. Database starts.
2. Reads WAL logs.
3. Replays committed transactions.
4. Removes incomplete transactions.

12. Connection Handling

Client connects to PostgreSQL server.
A backend process handles each connection.

Too many connections consume memory.

Solution:

Use connection pooling.

13. PostgreSQL vs MySQL

PostgreSQL:

- Process based
- Strong MVCC
- Advanced optimizer
- Object relational

MySQL:

- Thread based
- Depends on storage engine

14. Advanced Interview Questions

Why object-relational?

Supports objects, custom types and extensions.

Why rows are not updated directly?

Because PostgreSQL uses MVCC.

Why indexes?

To reduce disk scanning.

What is checkpoint?

A point where modified pages are flushed to disk.